Deep Learning for Road Network Extraction from Open-Access Low-Resolution Satellite Imagery in South African Geographies

Jing Yeh
University of Cape Town
Cape Town, South Africa
yhxjin001@myuct.ac.za

Kelvin Wei
University of Cape Town
Cape Town, South Africa
wxxkel001@myuct.ac.za

Catherine Li
University of Cape Town
Cape Town, South Africa
lxxcat004@myuct.ac.za

Abstract

Deep Learning (DL) has proven effective at automating road extraction from high-resolution aerial imagery. Road extraction from open-access, low-resolution satellite imagery is an emerging frontier. Promising DL methodologies, such as Geospatial Foundation Models (GFM) and data-centric approaches, can leverage the unique qualities of satellite data to monitor roads in diverse geographical regions in South Africa. This project addresses the gap in the application of DL road extraction to open-access satellite imagery. We will first curate a diverse South African road detection dataset to adapt and evaluate our approaches for satellite imagery. Afterwards, we will use the dataset for TerraMind GFM fine-tuning, SAM-Road GFM backbone adaptation, and data-centric augmentations using complementary modalities and super-resolution. To provide practical utility in assisting research into the impacts of roads, our methods will be integrated into an end-to-end pipeline.

1 Introduction

While deep learning (DL) has proven effective at automating road extraction from high-resolution imagery [23], its practical utility is constrained by the prohibitive costs and restrictive licensing of high-resolution data. Furthermore, current research exhibits a significant urban bias, producing models that fail to generalise to under-represented peri-urban and rural regions [37]. Open-access alternatives, such as OpenStreetMap (OSM) [36] and Sentinel satellite imagery [18] offer a promising and practical direction for road extraction research that reflects these underserved domains. However, road extraction using open-access satellite imagery remains an under-explored frontier. The complexities of processing satellite imagery with occlusion and low-resolution constraints present significant challenges in research. OSM presents additional challenges stemming from uneven coverage and outdated data, which compromise the accuracy of road labels.

Geospatial foundation models (GFM), a model-centric approach, are promising for leveraging vast volumes of unlabelled satellite imagery during pretraining and can be effectively adapted to downstream road extraction tasks through transfer learning. Recent advances include the multi-modal TerraMind [15] and multi-temporal Prithvi-EO-2 [35], which leverage unique qualities of satellite data.

Despite recent advances in model-centric approaches to remote sensing segmentation, data quality remains a primary determinant of performance. There is a growing body of work on low-resolution road segmentation, which focuses on improving the data inputs available for training rather than the model architecture itself. Two directions dominate: augmenting optical Sentinel-2 imagery with complementary modalities, such as Synthetic Aperture Radar (SAR) Sentinel-1 imagery [1, 22], and enhancing low-resolution images through Super-Resolution (SR) [1, 11, 27].

In this project, we propose DL approaches for road extraction using open-access satellite imagery. Our research primarily follows three phases: (1) the curation of a diverse South African dataset, (2) the evaluation of data-centric approaches by Jing, fine-tuning of the TerraMind GFM by Catherine, and the GFM backbone adaptation of the SAM-Road graph-based model by Kelvin; and (3) integrating the resulting models into InstaGeo [43], an end-to-end geospatial framework. The following sections describe the gaps motivating our research in Section 3. Our methodology is detailed in Section 4 and concludes with a project plan in Section 7.

2 Related Work

2.1 Satellite Data Modalities

Abundant open-access satellite imagery is made available by the Sentinel-1 and Sentinel-2 programs [18]. Recently, extracting temporal changes in road networks from satellite imagery has provided a basis for investigating the impacts of roads [34], particularly vital given that roads typically negatively impact ecosystems [28].

Sentinel-2 provides multispectral data (RGB, Red Edge, Near Infrared (NIR), Short-wave Infrared (SWIR)) at 10–60m resolutions [17]. Our approaches take advantage of combinations of multi-modal satellite imagery. Different bands are useful in distinguishing surface materials such as vegetation, water, and asphalt for land cover classification [9]. Sentinel-1 includes Synthetic Aperture Radar (SAR) imagery. SAR can penetrate cloud occlusion and canopy cover, which commonly block Sentinel-2 sensors [29]. Work from Lin *et al.* [22] and Ayala *et al.* [1] has shown that fusing SAR with optical input improves road segmentation accuracy.

2.2 Super-Resolution

Super-resolution (SR) is the task of reconstructing a high-resolution image from a low-resolution input. Aybar *et al.* [2] introduced SEN2SR, a SR framework specifically designed for Sentinel-2 imagery, capable of upsampling Sentinel-2 from 10m to 2.5m while maintaining spectral and spatial consistency. The framework is released with two main variants: SEN2SR-lite, a CNN-based model optimised for inference speed, and SEN2SR, a Mamba-based variant that achieves greater spatial consistency at significantly higher computational cost.

Two paradigms have been proposed for combining SR with road segmentation. Ayala *et al.* [1] use SR as an explicit preprocessing step before passing imagery to a segmentation network, finding that pre-upsampling improves segmentation accuracy but that bicubic interpolation outperforms learned SR for this purpose. Pereira and

dos Santos [27] propose an end-to-end framework that jointly trains an SR module and a segmentation network via a weighted joint loss. However, this approach requires high-resolution ground truth, which is unavailable for most African deployment contexts.

Haris *et al.* [11] propose a compound loss function that balances image reconstruction with task-specific optimisation. Their results show that fine-tuning the SR module on task gradients alone outperforms both low-resolution-input and SR-preprocessing baselines for object detection, despite reduced perceptual quality. We adapt this for road segmentation, since it requires only mask supervision rather than high-resolution imagery.

2.3 Road Segmentation Models

U-Net [33] is a common standard benchmark baseline model that utilises a symmetric encoder-decoder architecture with skip connections to preserve feature details. The encoder extracts feature information, and the decoder produces task-specific predictions. U-Net often utilises a ResNet backbone to benefit from transfer learning and residual connections to prevent the vanishing gradients problem. D-LinkNet [45] is a specialised model for road extraction. It integrates dilated convolutions in the central block to expand the receptive field. This architecture is most notably recognised for winning the CVPR DeepGlobe 2018 Road Extraction Challenge. DSCNet [30] introduces Dynamic Snake Convolution, a kernel that adaptively deforms to follow elongated, curved structures suited to thin road geometry.

2.4 Foundation Models

Foundational backbones are heavily utilised with transfer learning in computer vision tasks. The Segment Anything Model [20] (SAM) and SAM 2 [31], developed by Meta AI, is a general vision transformer (ViT) foundational model pre-trained on natural RGB images [32]. It understands general image features, but struggles to generalise well to satellite imagery [40]. SAM has been adapted for road segmentation tasks to address these issues [7, 8, 19, 26, 44].

GFMs trained on satellite imagery may better generalise to road extraction. Task-specific Earth Observation tasks can utilise the TerraTorch [10] framework to fine-tune GFMs.

Prithvi-EO-1 [14] and 2 [35] are GFMs trained on HLS data at 30m resolution with a vision transformer (ViT) backbone. Prithvi-EO-1 is a 100M parameter GFM trained on US-only data that accepts multi-temporal inputs. Prithvi-EO-2 expands on Prithvi-EO-1, with a 300M and 600M parameter version trained on global data that additionally supports location and temporal encodings, which the model learns to use. Specifically, Prithvi-EO-2 is designed to focus on long-term trends and seasonal patterns [35].

Terramind [15] is a GFM trained on various satellite imagery and modalities, with a ViT backbone. It is one of the few GFMs that simultaneously accept and output multiple satellite modalities (such as optical and SAR, text captions, geolocation coordinates) as complementary input [40]. Terramind introduces a novel recursive technique called "Thinking-in-Modalities" (TiM), which allows the model to generate and append synthetic modalities to the original input during inference.

2.5 Adapting Foundational Models

Adaptation is often required for Foundational models (FM) for better performance on a specific downstream task. Finetuning can be done by fully fine-tuning, or with parameter-efficient fine-tuning (PEFT) techniques to adapt FMs. Full fine-tuning for FMs is often resource-intensive, due to their large size. As a result, PEFT techniques are often explored. The essence of PEFT is to selectively update a subset of parameters from the pre-trained model, or a small set of new parameters injected into the model. Widely explored examples of these two PEFT approaches include Low-Rank Adaptation [13] (LoRA) and adapters, respectively. Additionally, researchers could modify more fundamental architectural choices of FMs for downstream tasks. Examples include changing the decoder to better leverage the encoder-generated features [4], or the backbone of the FM.

2.6 Graph-Based Models

Graph-based road network extraction models predict vectorised road networks of nodes and edges. Graph-based methods consist of iterative methods and one-shot methods. SAM-Road [12] is a one-shot graph-based method which utilises the SAM foundational model to predict road networks. The model obtained state-of-the-art performance on high-resolution datasets. It consists of a transformer-based graph neural network, which allows inference at significantly faster speeds than the state-of-the-art iterative RNGDET++ model [42].

3 Problem Statement

Current road extraction models are overwhelmingly trained on high-resolution, urban datasets [38]. Transitioning to freely available 10m satellite imagery introduces additional challenges in detecting sub-pixel features [21]. The use of low-resolution satellite imagery for road detection remains relatively unexplored and presents an opportunity for automated detection of road extraction across South African geographies.

Furthermore, existing datasets lack the geographic and biome diversity of peri-urban and rural South Africa, limiting model generalisability. Addressing this gap is helpful for organisations like the African Climate and Development Initiative (ACDI), interested in automated temporal road network analysis to isolate road impacts on landscape change. Currently, the lack of models tailored to African landscapes and low-resolution imagery directly impedes this downstream research.

3.1 Research Aims

This project aims to develop an end-to-end pipeline for extracting road networks from open-access 10 m satellite imagery, in urban, peri-urban and rural South African contexts. This aim is broken down into 3 phases:

(1) Curate a diverse South African road extraction dataset with accurate, high-quality labels suitable for training and evaluating DL models.
(2) Explore 3 DL approaches: data-centric with satellite image augmentation and super-resolution, fine-tuning approaches for GFM (TerraMind) and GFM backbone adaptation of graph-based model (SAM-Road).

(3) Integrate approaches into the InstaGeo pipeline, providing a deployable end-to-end road network detection tool for South African contexts.

The lack of diverse, local road extraction benchmarks motivates our focus on curating a representative South African dataset. By integrating our resulting approaches into an end-to-end pipeline, this project enables practical applications of the research outside of curated datasets.

3.2 Research Questions

Our research questions are divided into three distinct deep learning approaches, each tackled by a different team member. Data-centric methods focus on enhancing input quality to improve prediction accuracy. Meanwhile, GFM approaches leverage the domain-specific satellite representations learned by GFMs to better identify road structures. Finally, graph-based models are utilised for their inherent inductive bias, ensuring topological connectivity in the predicted networks.

3.2.1 Data Centric Approaches to Improve Road Segmentation on Open-Access, Low-Resolution Satellite Imagery - Jing.

- A1 How do additional input modalities (Sentinel-1 SAR, 20m optical bands) affect road segmentation performance at native 10m resolution?
- A2 How do deterministic upsampling, explicit super-resolution, and joint super-resolution training affect road segmentation performance on low-resolution satellite imagery, where roads often occupy sub-pixel widths?
- A3 Does combining the best-performing modality configuration with the best-performing resolution-enhancement strategy yield gains beyond either component alone?

3.2.2 GFM Backbone Adaptation in Graph-Based Road Network Extraction Model - Kelvin.

- B1 To what extent does adopting a GFM backbone improve the baseline road detection performance of SAM-Road on Sentinel-2 RGB imagery?
- B2 What performance benefits arise from leveraging the multimodal capabilities of GFMs in road extraction over RGB input?
- B3 To what extent do spatiotemporal embeddings improve the robustness of road extraction across satellite imagery subject to seasonal variation?

3.2.3 TerraMind GFM Adaptation - Catherine .

- C1 To what extent does a specialised decoder (DSCNet) improve TerraMind's road segmentation performance, compared to the decoders of general architectures used in the field (FCN, U-Net, UPerNet) and TerraMind's original decoder?
- C2 How do individually applied PEFT techniques (LoRA and, as an extension, an adapter) compare to fully fine-tuning TerraMind?
- C3 Which combination of the implemented decoder architecture and PEFT techniques (LoRA and potentially AdaptFormer) from the above questions yields the best segmentation performance?
- C4 To what extent do large models (TerraMind's zero-shot performance and its best-performing variant from the above experiments) differ in performance relative to the medium-sized models (U-Net and D-LinkNet)?

4 Methodology

Individual sections adopt a waterfall approach. For shared sections 4.1, 4.3 and 4.7, we adopt lightweight, agile practices where appropriate, tracked via a Kanban board and GitHub feature-branch workflows. We start with a minimal viable prototype and iteratively improve on top of existing work. We host weekly team and supervisor synchronisation meetings to ensure alignment.

4.1 Data Curation

Our regions of interest are distributed across rural, peri-urban and urban areas in different biomes in South Africa to obtain a representative road extraction dataset. The primary dataset outputs are road masks, graph labels and the corresponding satellite imagery. Following an iterative approach, we first create a prototype that involves manual processes and scripts for a single region. More regions and better processes are then iteratively incorporated.

Curating the dataset requires several steps: data aggregation, semi-automated processing, manual inspection and categorisation. We utilise the Google Earth Engine cloud platform to preprocess and obtain satellite imagery. The semi-automated process involves overlaying road vector data, such as OpenStreetMap (OSM), onto satellite imagery. These vectors are corrected for inaccuracies before being converted into road masks and graph labels. Categorisation includes splitting data into train, validation and test sets, with stratification across urbanisation categories via Jenks Natural Breaks [16] on road-pixel density.

4.2 Evaluation Protocol

To quantify model performance, we utilise pixel-based metrics: Precision, Recall, F1-score, and Intersection over Union (IoU) [25]. These metrics offer a more representative evaluation, particularly given the class imbalance between road and background pixels.

However, pixel-based metrics do not capture the topological correctness of road networks, as demonstrated by Van Etten *et al.* [5]. To address this limitation, we supplement our evaluation with two graph-based metrics: Average Path Length Similarity (APLS), which measures the dynamic similarity based on the shortest distance of node pairs, and TOPO, which measures the structural similarity of the road networks. Calculating graph metrics (TOPO and APLS) from raster masks introduces some conversion noise. We accept this as systematic across methods and therefore not biasing the comparisons between them. All experiments will follow the above-described evaluation methodology, unless stated otherwise.

4.3 Baselines

We evaluate our curated dataset using U-Net [33] as a general segmentation baseline and D-LinkNet [45] as a specialised baseline model designed for road extraction. Simple augmentation strategies, such as rotation and scaling, would be applied to assist generalisation performance. A simple hyperparameter grid search of

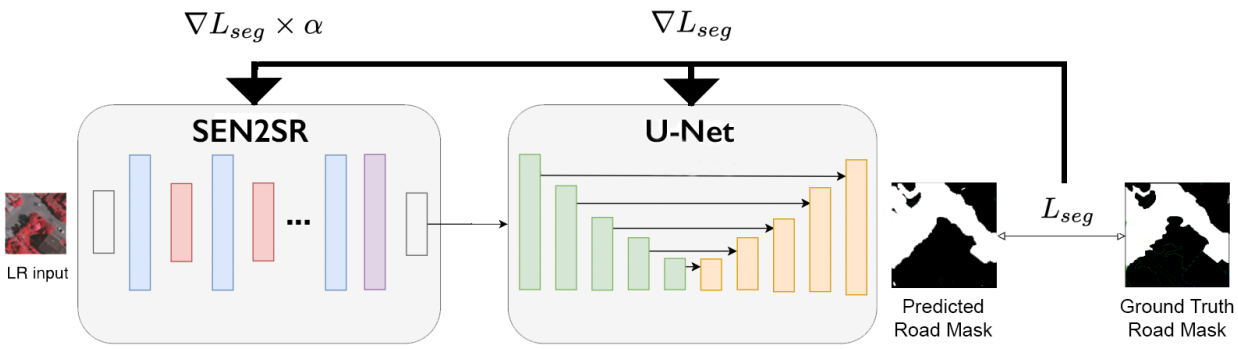


Figure 1: Joint SR and road Segmentation architecture on segmentation loss gradient (∇L_{seg}) alone, with α scaling the SR module's learning rate. Base figure components adapted from Pereira and dos Santos [27].

important parameters would be utilised to tune parameters like the dilation rate of the D-LinkNet, which controls the receptive field.

A comprehensive evaluation of our individual approaches (Sections 4.2, 4.5, 4.6) will be conducted on the test set against our baselines. We train each best-performing configuration with three random seeds to estimate training variance, with results reported as mean ± std. For pairwise statistical comparisons between configurations we use Wilcoxon signed-rank tests [39] on per-tile metric values, treating each test tile as a paired observation. We additionally report 95% confidence intervals on between-model performance differences via paired bootstrap over test tiles, providing a magnitude estimate alongside significance.

4.4 Data-Centric Approaches

4.4.1 Modality Investigation (RQ A1). The contribution of additional modalities is evaluated as a sequential ablation rather than an exhaustive grid (configurations M0-M3 in Table 5). The baseline uses Sentinel-2 RGB and NIR (M0). Two augmentations are tested independently and in combination: Sentinel-2 20m bands (B5-B7, B8A, B11, B12) bicubically upsampled to 10m and concatenated as additional input channels (M1); and Sentinel-1 SAR, despeckled and integrated via channel-wise stacking with the optical input, following Ayala *et al.* [1] (M2). Bicubic interpolation is the default upsampling for the 20m bands; identifying the optimal sharpening strategy is outside the scope of this work. If time permits, an ablation will assess whether explicitly computed spectral indices (e.g., Normalised Difference Vegetation Index, Normalised Burn Ratio) provide additional value beyond the raw 20m bands, given that these indices are derivable from the included bands.

4.4.2 Resolution Enhancement (RQ A2). The second research question compares three strategies for producing 2.5m output, supervised by high-resolution masks rasterised from OpenStreetMap onto each Sentinel-2 tile's grid: (i) bicubic interpolation of the input to 2.5m as a deterministic baseline; (ii) explicit super-resolution using SEN2SR [2] as a frozen preprocessing step, selected for its reported spectral and spatial consistency; and (iii) joint training, inspired by Pereira and dos Santos's framework [27] for combined SR and segmentation training. Their approach uses a joint loss combining L1 reconstruction (against HR ground-truth imagery) and cross-entropy (against HR masks). Due to the absence of co-located Sentinel-2 and high-resolution imagery for our deployment regions, we adopt a segmentation-only variant inspired by Haris *et al.*'s TDSR-DET [11]: a pretrained SEN2SR is fine-tuned alongside the U-Net via backpropagation of segmentation loss gradient alone, omitting the reconstruction term. Differential learning rates are tuned to prevent destruction of pretrained SR weights early in training, as shown in Figure 1. This variant additionally tests whether segmentation gradients alone are sufficient to adapt pretrained SR weights for the downstream task.

4.4.3 Combination (RQ A3). The strongest configuration from RQ A1 and the strongest strategy from RQ A2 are integrated and compared against each component individually. This bounded scope avoids the combinatorial expansion of testing all modality-resolution pairs.

4.5 Geofoundation Backbone in Graph-Based Models

4.5.1 Experimental Framework: SAM-Road Model Control. The following experiments will utilise the SAM-Road [12] model for vectorised road network prediction. The model was chosen due to its architectural structure, which leverages a foundational ViT backbone and state-of-the-art performance. By holding SAM-Road's geometry and topology decoders constant (see Figure 2), the effects of different encoder backbones can be isolated. Table 6 details the experiments.

SAM-Road was originally trained and evaluated on high-resolution datasets, which are <1m resolution. Sentinel-2 RGB imagery has a spatial resolution of 10m. To optimise hyperparameters for low-resolution imagery, experiments utilise Bayesian Optimisation with Hyperband (BOHB) [6], tracked via Weights and Biases Sweeps. BOHB efficiently combines Hyperband to systematically stop underperforming trials with a Bayesian optimiser that uses validation losses from finished runs to probabilistically select improved parameters for subsequent trials.

To evaluate the predicted road network graphs, the TOPO and APLS metrics will be calculated for experiments. Furthermore, to enable comparison with the pixel-based baseline models described in Section 4.3, the vector graphs of the best-performing model will

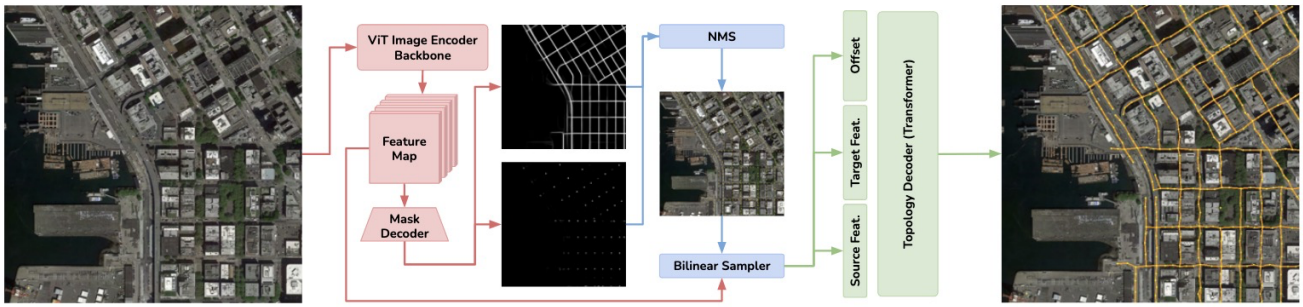


Figure 2: SAM-Road Architecture [12]

be rasterised into binary road masks by applying a fixed-width overlay to the predicted graphs.

4.5.2 GFM Backbone Investigation. The first research question compares 4 different foundation models. The default ViT encoder in SAM-Road will be systematically swapped with geospatial encoders, specifically TerraMind [15], Prithvi-EO-2.0 [35], and Prithvi-EO-1.0 [14]. The Terratorch framework will be utilised to easily integrate these GFM backbones, employing intermediate adapter layers (necks) to resolve any architectural incompatibilities.

To establish a valid comparative baseline, this primary evaluation will be strictly restricted to Sentinel-2 RGB bands across the curated South African dataset, utilising APLS and TOPO metrics to assess performance. The original SAM model will serve as the baseline architecture. However, because SAM's performance on low-resolution remote sensing data is known to be limited [40], all foundational model encoders will undergo full fine-tuning during training to ensure a fair comparison.

4.5.3 Multimodality Investigation. TerraMind and Prithvi inherently support multimodal data processing. Consequently, the secondary question investigates the performance impact of incorporating Sentinel-1 SAR and multi-spectral (10m-30m) bands. Bands that are not 10m would be upsampled to 10 m with bicubic interpolation. Prithvi-EO-1.0 is not explicitly trained on SAR; a composite SAR representation will replace one of its existing input channels based on results in Section 4.2.

4.5.4 Spatiotemporal Investigation. Prithvi-EO-2.0 and TerraMind natively utilise location and temporal embeddings to process satellite imagery. To evaluate the impact of these embeddings on road extraction, an intra-model comparison will be conducted, benchmarking the respective models against their spatiotemporal-embedded variant. These embeddings potentially allow the models to interpret seasonal variations and environmental shifts (such as vegetation changes). Therefore, we will evaluate model predictions across distinct time periods (summer and winter) to determine the robustness of the extraction models to seasonal variations in South Africa.

4.6 TerraMind GFM Adaptation

Refer to Table 7 in the Appendix for the specific experiments. All bands available in the curated dataset will be used as input in every experiment. The experiments will be based on [4], which explored LoRA [13], an adaptor, as well as various decoder architectures. All their evaluation techniques are available via the open-source Python package TerraTorch. The Python package also contains the TerraMind Model, which will be used with substantial hyperparameter tuning. The paragraphs below give an overview of each component in this approach, with explicit programming requirements specified when applicable. To isolate the effects of PEFT techniques, the linear decoder will be used for RQ C2.

4.6.1 Decoder (RQ C1). The decoder to be evaluated includes the linear, the original TerraMind decoder [15], FCN [24], UPerNet [41], U-Net [33], and DSCNet [30]. The linear decoder is the control (as in [4]), the TerraMind decoder assesses TerraMind's inherent ability to generalise to road segmentation, and the other decoders have distinct properties described in Section 2 that may be helpful for road extraction. The TerraMind encoder will be fully fine-tuned with each of the listed decoder heads, then evaluated. The first five decoders are integrated into the Python library TerraTorch [10]. The DSCNet decoder will require code extraction from [30]'s GitHub, explicit registration within TerraTorch and adaptations for compatibility.

4.6.2 PEFT Techniques (RQ C2). LoRA [13] introduces pairs of trainable low-rank weight matrices in pre-trained transformer models that are absorbed into the existing architecture at runtime. The hyperparameters configurable specific to LoRA include the target modules for injecting LoRA, rank, and alpha (a scaling factor). The latter two hyperparameters will be tuned. As no specific parameter budget will be imposed, LoRA will be injected into each transformer layer in TerraMind. As an extension, the different target modules could be systematically varied along with the other two hyperparameters.

A further extension is exploring adaptor fine-tuning by implementing the AdaptFormer framework [3] in the context of TerraMind. This will require a deep understanding of the model and some programming. The framework is chosen because it inserts adapters in the Multi-Layer Perceptron of transformers, which remain consistent across ViT backbones, instead of the variable attention module. Example applications of AdaptFormer in road segmentation include [8].

4.6.3 Ablation (RQ C3). The best performing decode, along with the LoRA and the adaptor (if implemented), will be integrated

together, which is possible as they are orthogonal to each other [13]. The decoder, LoRA and the adaptor are then systematically ablated and compared. This is to observe any interaction effects. Specifically, to empirically evaluate whether the different ways of combining the techniques in RQ1 and RQ2 will substantially or marginally improve model performance.

4.6.4 Baselines Evaluation (RQ C4). The best resulting TerraMind variant from the RQ C3 is compared against TerraMind's zero-shot performance and the baselines described in Section 4.3.

4.7 Integration Into InstaGeo Framework

InstaGeo [43] is an end-to-end, open-source framework that retrieves data for model inference and visualises predictions. The framework currently supports similar geospatial machine learning tasks, such as crop segmentation. The framework has an interactive web-map application that can be adapted to visualise road predictions. After the team's individual components, the best resulting approach from each team member will be integrated into the framework, along with road visualisation capabilities.

5 Ethics

This project requires no formal ethics clearance as it relies on publicly available, low-resolution satellite imagery and road annotations where individuals cannot be discerned. No human interaction or personal data analysis is involved. All utilised models and software are open-source or academic-licensed.

6 Anticipated Outcomes

This project aims to contribute to the mentioned research gaps while delivering a practical road extraction tool:

(1) An annotated, multimodal road extraction dataset specific to South Africa.
(2) Empirical guidance on which auxiliary modalities and resolution-enhancement strategies yield meaningful improvements.
(3) Insights into the efficacy of GFM backbones for graph-based topological road extraction and the impact of spatiotemporal embeddings.
(4) The performance comparisons of various decoder architectures and Parameter-Efficient Fine-Tuning (PEFT) techniques for GFMs.
(5) A deployable inference pipeline integrated into the InstaGeo framework, to enable downstream road analysis, such as road network changes, for ACDI research.

Success requires the best model from each approach to demonstrate consistent performance (via standard deviations and confidence intervals) and outperform baselines. Otherwise, underperformance is analysed. A design challenge will be adapting InstaGeo's visualisations for non-technical users.

7 Project Plan

7.1 Resources Required

7.1.1 Data Resources. InstaGeo obtains satellite imagery bands from a direct source. Satellite imagery is batch preprocessed and obtained via Google Earth Engine (academic license). Road annotations are sourced from OSM and Overture.

Table 1: Project Deliverable and Due Dates

Deliverable	Due Date
Project Proposal Demo	28 April
Final Proposal	5 May
Project Progress Demo	27 July
Final Paper Scaffold	10 August
Full Draft Report	24 August
Final Project Report	4 September
Project Code	14 September
Project Demo	16 September
Poster	28 September
Website	5 October

7.1.2 Hardware Resources. Computational experiments, such as GFM tuning and data processing, will require access to UCT's high-performance computing (HPC) facilities. Additional storage drives are needed to house the data. Google Colab and Kaggle GPU resources are used for prototyping before using the UCT HPC facilities.

7.1.3 Software Resources. Google Earth Engine is used for batch semi-automated processing. Manual annotation can be done with GIS software like QGIS. Weights and Bias platform for logging training and hyperparameter metrics. Python libraries such as TerraTorch [10] and PyTorch will be used for building and training models. Open source code on GitHub will be used during the exploration of different methodologies and InstaGeo framework.

7.2 Risks

Refer to Table 3 for the tabulated risks and their corresponding mitigation, monitoring, and management strategies.

7.3 Deliverable

Refer to Table 1 for the associated deliverables of the project.

7.4 Milestones

Refer to Table 2 for the associated milestones of the project.

7.5 Timeline

Refer to the Gantt chart in Figure 3 for the timeline visualisation with dates.

7.6 Work Allocation

Refer to Table 4 for work load division.

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Table 2: Project Milestones and Completion Dates

Milestone	Due Date
Project Proposal slides Done	27 April
Project Proposal Finalised	5 May
Dataset Curated	19 May
Model Baseline Curated + Evaluated	26 May
Individual methodologies Implemented	7 July
Methodology Evaluation Complete	21 July
Project Progress slides Done	27 July
Integration into InstaGeo Pipeline	4 August
Final Paper Scaffold Done	10 August
Full Draft Report Done	24 August
Final Project Report Done	4 September
Project Code Completed	14 September
Project Demo Slides Done	16 September
Poster Done	28 September
Website Done	5 October

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A AI Usage Statement

Generative AI (Claude, ChatGPT, Gemini) models were used for reducing unnecessary redundancy, fixing LaTeX compilation bugs, and improving grammar. AI Feedback on the clarity of content was incorporated into the writing. All technical methodologies, research questions, and literature analyses remain original.

Table 3: Risk Assessment and Mitigation Plan

Risk	Probability	Severity	Mitigation	Monitoring	Management
Data curation takes too long	Medium	High	Automate preprocessing pipeline; begin data curation during proposal phase	Track dataset size against weekly milestones	Reduce dataset scope or begin methodology on a sampled subset
Only a small amount of high-quality data is curated	High	High	Apply data augmentation (rotations, flips, cropping).	Monitor model performance early on subsets	Adjust scope to proof-of-concept rather than performance
HPC usage is delayed	High	Medium	Submit jobs incrementally as components are ready; avoid last-minute submissions	Track job runtimes to inform planning	Develop locally on CPU; scale experiments using GPU on Kaggle/Colab
Class imbalance in road segmentation	High	Low	Use metrics such as IoU, mIoU, F1-score, ALP; apply specialised losses (e.g., clDice)	N/A	Same as mitigation
Insufficient time for the project scope	High	High	Define minimum viable scope and prioritise core experiments	Compare progress against weekly milestones	Reduce scope where necessary
Insufficient expertise delays completion	High	High	Consult literature, tutorials, and existing implementations; seek supervisor guidance	Track time spent blocked on tasks	Simplify tasks or reuse/adapt existing solutions

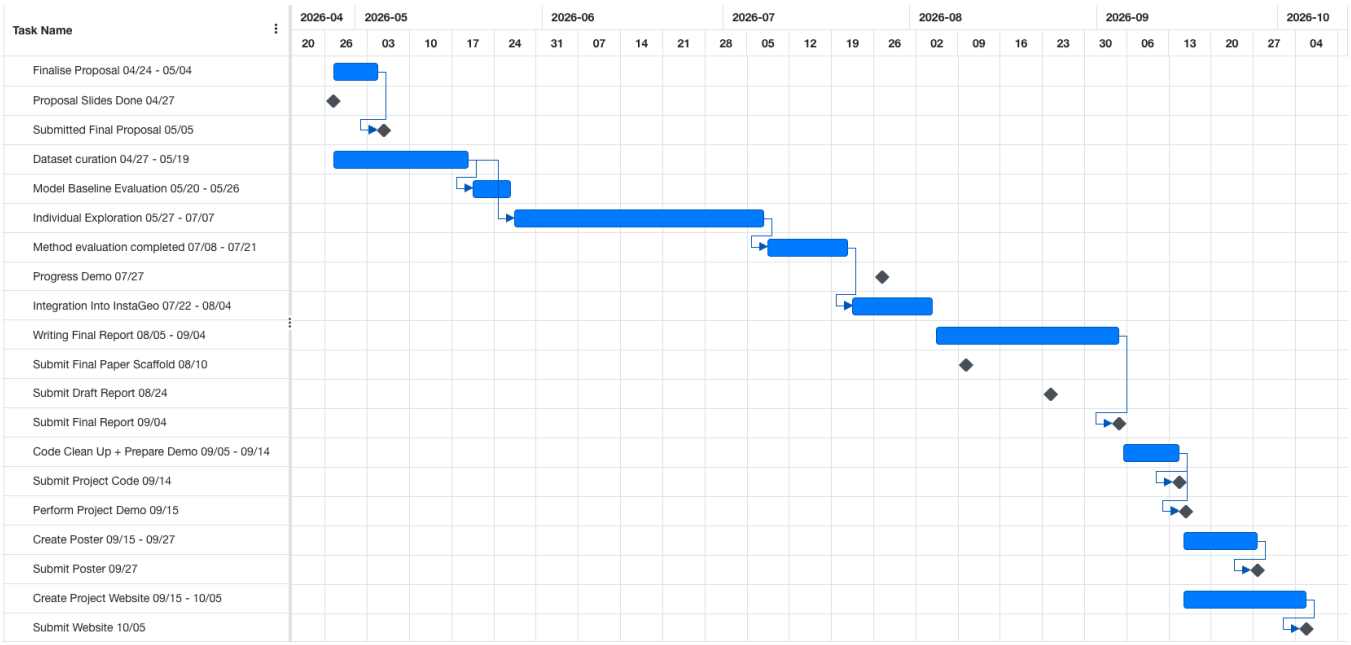


Figure 3: Gantt Chart of Project

Table 4: Group Work Allocation. Refer to the timeline section for the task description.

Task	Role
Data Curation	All group members will work on curating the dataset using an adapted agile approach described in Section 4 and 4.1.
Baseline Evaluation	Extension of data curation and baselines will be compared against our individual approaches. All group members will work together using the agile approach described in 4.
Individual Approach	Team member will work on their respective individual approaches, which will mostly consist of a waterfall approach.
InstaGeo Integration	Team members will be responsible for the integration of their respective models. The UI integration and coordination will be worked on together using an agile approach.

Table 5: Experiments for the data-centric approach

ID	RQ	Configuration	Input	Resolution Strategy	Output	Architecture
M0	–	Baseline	S2 RGB+NIR (4ch)	None	10m mask	U-Net + ResNet-34
M1	A1	+ 20m bands	S2 RGB+NIR + 20m bands (10ch)	Bicubic to 10m	10m mask	U-Net + ResNet-34
M2	A1	+ SAR	S2 RGB+NIR + S1 SAR (6ch)	None	10m mask	U-Net + ResNet-34
M3	A1	+ 20m bands + SAR	S2 RGB+NIR + 20m + S1 SAR	Bicubic to 10m for 20m	10m mask	U-Net + ResNet-34
R0	A2	Bicubic upsampled input	S2 RGB+NIR	Bicubic to 2.5m	2.5m mask	U-Net + ResNet-34
R1	A2	Explicit SR (frozen)	S2 RGB+NIR	SEN2SR preprocessing	2.5m mask	U-Net + ResNet-34
R2	A2	TDSR-DET (joint SR + seg)	S2 RGB+NIR	SEN2SR fine-tuned	2.5m mask	SEN2SR + U-Net (ResNet-34)
C1	A3	Combined (best of RQ A1 + RQ A2)	Per RQ A1 winner	Per RQ A2 winner	2.5m mask	Per RQ A2 winner

Table 6: Experiments for the GFM backbones for graph-based road detection models

ID	RQ	Configuration	Input	Backbones
G0	B1	RGB	S2 RGB	TerraMind, Prithvi-EO-1.0+2.0, SAM
G1	B2	Multi-modal	S2 RGB+NIR + 20m bands + S1 SAR	TerraMind, Prithvi-EO-1.0+2.0
G2	B3	Spatiotemporal embeddings	Seasonal S1 & S2 Bands + spatiotemporal embeddings	TerraMind, Prithvi-EO-2.0

Table 7: Experiments for Different fine-tuning techniques. The bolded experiment IDs are the core experiments, while the rest are the extensions.

ID	RQ	Fine-Tuning Strategy	Architecture
PLAIN_0	-	Full fine-tuning	TerraMind Encoder + Linear Decoder
PLAIN_0	C1	-	-
DECODER_1	C1	Full fine-tuning	TerraMind Encoder + FCN Decoder
DECODER_2	C1	Full fine-tuning	TerraMind Encoder + U-Net Decoder
DECODER_3	C1	Full fine-tuning	TerraMind Encoder + UPerNet Decoder
DECODER_4	C1	Full fine-tuning	TerraMind Encoder + DSCNet Decoder
DECODER_5	C1	Full fine-tuning	TerraMind Encoder + TerraMind Decoder
PLAIN_0	C2	-	-
PEFT_1	C2	LoRA [13] (Rank, and Alpha hyperparameter tuning)	TerraMind Encoder + Linear Decoder
PEFT_2	C2	LoRA [13] (Rank, Alpha, and Target Module hyperparameter tuning)	TerraMind Encoder + Linear Decoder
PEFT_3	C2	Adaptformer [3]	TerraMind Encoder + Linear Decoder
PLAIN_0	C3	-	-
ABLATION_1	C3	Best from 2 + Best from 3	TerraMind Encoder + Best decoder from 1
ABLATION_2	C3	Best from 2	TerraMind Encoder + Best decoder from 1
ABLATION_3	C3	Best from 3	TerraMind Encoder + Best decoder from 1
ABLATION_4	C3	Best from 2 + Best from 3	TerraMind Encoder + Linear Decoder
ABLATION_5	C3	Best from 2	TerraMind Encoder + Linear Decoder
ABLATION_6	C3	Best from 3	TerraMind Encoder + Linear Decoder
ABLATION_7	C3	Full fine-tuning	TerraMind Encoder + Best decoder from 1
BASELINE_0	C4	No fine-tuning	TerraMind Encoder + TerraMind Decoder
BASELINE_1	C4	best strategy from 4	TerraMind Encoder + best strategy from 4
BASELINE_2	C4	Full fine-tuning	U-Net
BASELINE_3	C4	Full fine-tuning	D-LinkNet